Course Name: Database Management System

Course Outcome

CO1: Understand the concept of database management systems and Relational database.
CO2: Identify the various data model used in database design.
CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
CO4: Create and populate a RDBMS for a real-life application, with constraints and keys using SQL.
CO5: Select the information from a database by formulating complex queries in SQL.
CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.

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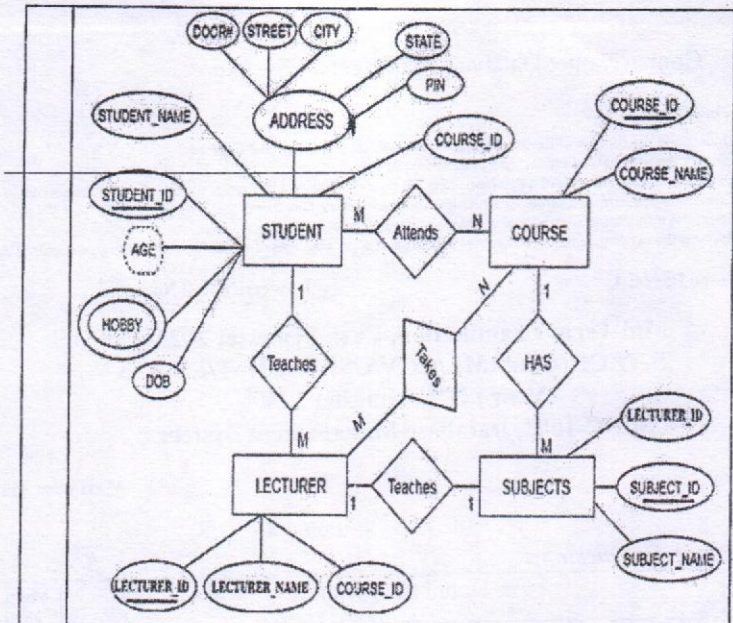
Mid Term Examination, Even Semester 2023-24
B.TECH(CS/AIML/CCV/CSF/DA/IIoT/EC-CS)
Year : II[nd], Semester : IV[th]
BCSC-1003, Database Management System

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Retail company needs to store and manage vast amounts of customer data, including purchase history, personal information, and preferences. They also need to quickly retrieve specific customer details for personalized marketing campaigns. In this scenario, which system would be more suitable, a database or a file system? Explain your choice and justify the advantages?	3	1	R	F
2	Consider a scenario in which you are appointed as a database administrator in an organization. What roles and responsibilities you are supposed to perform? Explain it.	3	1	A	C
3	Analyze the ER Diagram given in Figure 1 and reduce it to relational database schema. Identify Candidate, Primary & Foreign Keys and also determine the total numbers of Tables required.	3	3	A	P

	 Figure 1: ER Diagram				
4	Consider the Relation R(WXYZ) and Functional Dependencies given below: FD: {W → X, X → Y, Y → W}. Identify Candidate Key(s) in this relation.	3	6	U	C
5	Consider a database for student records containing fields like Student ID, Name, Address, and Phone Number. However, it's observed that some students share the same address. Explain the concept of normalization and identify which normalization form would be most suitable to address this redundancy.	3	6	U	C

Section – B

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Consider the following schema: DEVELOPER (**DEV_ID**, DEV_NM, ADDRESS) MANAGER (**MG_ID**, MG_NM, MG_PROJ_NM) COMPANY (**DEV_ID, MG_ID**, COMP_NAME) Attributes written in bold letters are prime attribute(s). Write Relational Algebra expression for the following queries: (a) Find the name of developers who live in 'Chennai'. (b) Find the name of Manager who is working on Project 'ABC'. (c) Find the name of manager who is managing Developer 'Arvind' for project 'XYZ'. (d) Insert a new tuple into relation MANAGER. (e) Delete records of Developers whose address is 'Pune'.	5	3	A	P
7	Consider the following relation: ***STUDENT*** *(R_NO, S_NAME, S_COURSE, CPI, ADDRESS, EMAIL_ID)* Take datatype and range as per the suitability of this question. Write the SQL syntax for following queries: (a). Write the syntax to create STUDENT table. (b). Insert a tuple in STUDENT table. (c). Delete the record of Student who live in 'Agra'. (d). List the details of the students in order of the descending of CPI. (e). List the student names those are starting with 'A' and with five characters.	5	4	A	P

8	Draw an ER Diagram for a "Hospital Management System". You need to keep track of Patients, Doctors, Medical Staff, and Departments. Patients are admitted to the Hospital, Doctors treat Patients, and each Department specializes in certain medical fields. Also identify the Primary Key, Foreign Key, Candidate Key used.	5	2	A	C